Battery Contribution and Battery Benefit — how the numbers in your case are produced

For the example you sent, this note explains how each figure in the result panel was produced and what it includes. It follows your own scenario throughout: **800 kW battery**, four modes — Transit 2 500 h · DP 1 000 h · Port 500 h · Manoeuvring 500 h.

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1. What the four columns are

Every row of an allocation table is one load, and the columns hold one calculation:

Column	Symbol	Meaning
Variation	**H**	the swing this load presents — the amount that has to be covered
Battery used	**I**	how much of the battery budget this row is allocated
Covered	**J**	`I × CoverageFactor` — the part of the swing the battery is credited with removing
Uncovered	**L**	`H − J` — what the engines still have to hold as spinning reserve

The identity `J + L = H` holds on every row and on every total. The battery never reduces the swing; it only moves who carries it.

2. How Variation (H) is derived

Three different formulas, depending on the load's function:

```
Peak shaving loads    H = average load × variation factor
Mission consumers     H = full rating                        (on or off, no percentage)
Class reserve (DP)    H = the stated requirement             (taken from the input directly)
```

Applied to your case:

```
Transit propulsion    11 463 × 1.10 (sea margin)  = 12 609.3   → × 5 %  = 630.5
Transit hotel         3 800                                    → × 2 %  =  76.0
Mission consumer      250 kW rating                                     = 250.0
DP redundancy         stated in the input                               = 800.0
```

The variation factors (propulsion 5 %, hotel 2 %) are configuration values, not derived.

3. How the budget is allocated

The battery is a fixed budget, allocated top-down in a fixed priority order:

```
DP reserve  →  DP demand  →  mission  →  propulsion  →  hotel
```

Each row takes what it asks for, or whatever is left, whichever is smaller. When the budget is exhausted the remaining rows take **zero** — they are not skipped or ignored, they simply arrive after the money is gone. That is the mechanism behind every `0` in the *Battery used* column.

4. Coverage factors

Function	Factor	Why
DP redundancy reserve	**1.00**	a class requirement is met or not met; 800 kW of battery satisfies 800 kW of requirement exactly
DP demand	0.50	
Mission consumer	0.50	
Propulsion	0.35	
Hotel	0.05	

All five are held in configuration and can be retuned. They are the reason *Battery used* and *Covered* are different numbers on every row except the DP reserve.

5. Transit — the full allocation

Total demand for battery power is `250 + 630.5 + 76 = 956.5 kW` against a budget of **800 kW**.

Row	H (variation)	I (used)	J (covered)	L (uncovered)
Mission (crane)	250.0	250.0	250 × 0.50 = **125.0**	125.0
Propulsion	630.5	**550.0**	550 × 0.35 = **192.5**	438.0
Hotel	76.0	**0**	0	76.0
	956.5	800.0	317.5	639.0

Mission is served first at its full 250 kW. Propulsion is next and receives the remaining 550 kW — 80 kW short of what it asked for. Hotel is last and the budget is empty.

6. DP — the full allocation

Total demand is `800 + 250 + 70 = 1 120 kW`, again against 800 kW.

Row	H	I	J	L
DP redundancy reserve	800.0	**800.0**	**800.0**	0
DP demand	0	0	0	0
Mission	250.0	**0**	0	250.0
Hotel	70.0	**0**	0	70.0
	1 120.0	800.0	800.0	320.0

The class reserve is first in the queue and is the one row covered 1:1, so it consumes the entire budget on its own. Everything below it receives zero.

7. The three rows you marked

Transit · Mission · 250 / 250 / 125 / 125. A mission consumer's variation is its full rating, not a percentage of it. The battery covers it in full, and 50 % of that counts as covered — hence 125 covered and 125 remaining on the engines.

Transit · Hotel · 76 / 0 / 0 / 76. Hotel is last in the priority order and the 800 kW was consumed by mission (250) and propulsion (550). Its full 76 kW therefore stays with the engines. Note that even if it had been served, the 5 % hotel factor would have credited only 3.8 kW of it.

DP · Mission · battery used 0. The 800 kW class reserve outranks it and takes the whole budget. With a larger battery the mission row would be the next one served.

8. Where the uncovered power goes

Uncovered reserve is added to the demand of the side that owns the load:

```
Transit   propulsion' = 12 609.3 + 438.0               = 13 047.3 kW   (shaft side)
          hotel'      =  3 800    + 125.0 + 76.0        =  4 001.0 kW   (switchboard)
DP        thrust'     =  1 000    +    0                =  1 000.0 kW
          hotel'      =  3 500    + 250.0 + 70.0        =  3 820.0 kW
```

The mission consumer's uncovered 125 kW lands on the **hotel** side — it is an electrical consumer on the switchboard, not a load on the shaft.

With the shaft generator at 500 kW, the Transit auxiliaries carry `4 001 − 500 = 3 501 kW`, and the main engine carries `13 047.3 + 500 = 13 547.3 kW`, i.e. **90.3 %** of its 15 000 kW rating.

The header figures are hours-weighted averages over all four modes, not sums: the main engine's `39 868 163 kWh / 4 500 h = 8 860 kW`.

9. The two tiles

```
Peak Shaving       = Σ J  over peak-shaving rows only     Transit 317.5 kW
Spinning Reserve   = Σ L  over all rows                   Transit 639.0 kW  ·  DP 320.0 kW
```

One display point worth flagging: in the DP table the *Covered* total reads **0** although the reserve row shows 800. The total is the Peak Shaving figure, and a class reserve is deliberately excluded from peak shaving — it is a different function, not a peak being flattened. The 800 kW is real and is fully accounted for in the Battery Benefit below. We agree the column reads like a broken sum and will relabel it.

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10. Battery Benefit (vs no battery)

This is a separate question from the integration levels. The levels ask what better control would save on the vessel as configured; Battery Benefit asks what **the battery itself** saves against the same vessel without one.

The full pipeline is run twice per battery-relevant mode, and both runs are independently optimised — the engine combination is re-selected in each world, so it is a like-for-like comparison:

```
World A (as configured)   demand = average + L        engines carry only the uncovered part
World B (no battery)      demand = average + H        engines carry the entire swing
Benefit = (FOC_B − FOC_A) × mode hours,  summed over the modes where the battery is active
```

For your case the two worlds differ as follows:

Mode	World A	World B	Extra load in World B	Hours
Transit	prop 13 047.3 · hotel 4 001.0	prop 13 239.8 · hotel 4 126.0	+192.5 · +125.0	2 500
DP	thrust 1 000 · hotel 3 820.0	thrust 1 800 · hotel 3 820.0	+800.0	1 000

In World B nothing is covered, so every row's full variation is added instead: Transit propulsion gains back its 192.5 kW and the mission consumer its 125 kW; DP gains back the whole 800 kW class reserve.

Result: 379.9 t/yr, or $296 318/yr.

DP contributes the larger part despite having 40 % of the Transit hours, because 800 kW of avoided reserve is a far bigger difference than 317.5 kW of shaved peak. The difference is not simply linear in kW: carrying the extra load can force an additional generator to start, and the engine then runs at a load point with a worse specific consumption, so the saving is taken from the optimised fuel figures of the two worlds rather than from a flat multiplication.

A caveat about reproducing this in the interface. Unticking *Enable Battery* does not give you World B. With no battery configured, the tool models the plain average load with the swing carried by nobody at all — which burns less than the battery case and makes the battery appear counterproductive. World B, where the engines genuinely carry the full swing, exists only inside the benefit calculation.

11. If more of the load should be covered

The battery is the binding constraint in both modes: Transit asks for 956.5 kW and DP for 1 120 kW, against 800 kW installed. At 800 kW the model serves the highest-priority rows in full and reports the remainder honestly as uncovered rather than spreading the budget thinly across all of them.